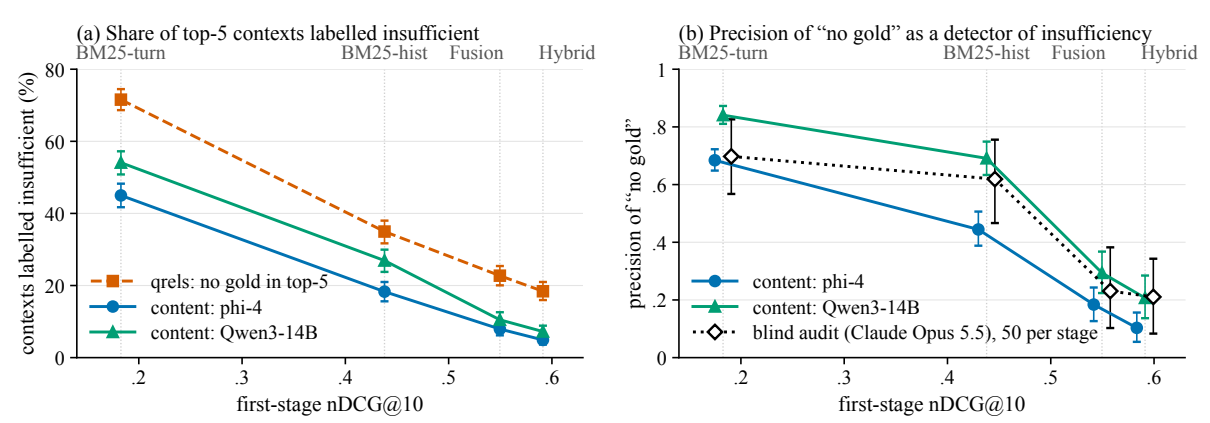
(a) Share of top-5 contexts labelled insufficient
BM25-turn
BM25-hist
Fusion
Hybrid
contexts labelled insufficient (%)
80
60
40
20
0
qrels: no gold in top-5
content: phi-4
content: Qwen3-14B
.2
.3
.4
.5
.6
first-stage nDCG@10
(b) Precision of "no gold" as a detector of insufficiency
BM25-turn
BM25-hist
Fusion
Hybrid
precision of "no gold"
1
.8
.6
.4
.2
0
content: phi-4
content: Qwen3-14B
blind audit (Claude Opus 5.5), 50 per stage
.2
.3
.4
.5
.6
first-stage nDCG@10